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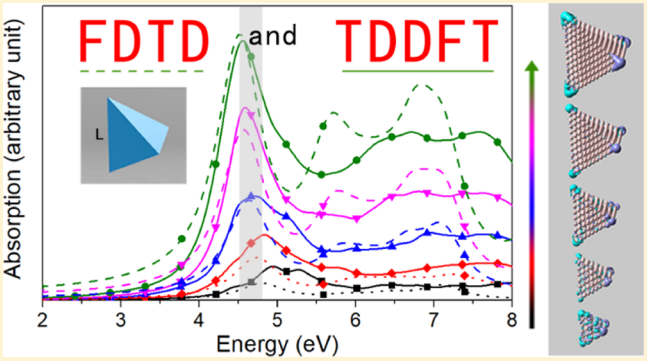
Size-Dependent Optical Properties of Aluminum Nanoparticles: From Classical to Quantum Description

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Supporting Information

ABSTRACT: The absorption spectra of a series of tetrahedral-shaped aluminum nanoparticles (ANPs) with side lengths ranging from L = 1.4 to 141.4 nm have been calculated. The size-dependent evolution of structures and spectra has been demonstrated. The plasmon resonance of ANPs is highly sensitive to the particle size and spans a wide spectral region from ultraviolet to visible. As L increases from 70 to 141 nm, an apparent spectral red shift is still observed, although their spectral line shapes do not change much. For the small clusters Al_n with $n \leq 560$ (or with $L \leq 3.68$ nm), their absorption spectra have been calculated by both the real-time time-dependent density functional theory (RT-TDDFT) scheme and the finite difference time domain (FDTD) classical electrodynamics method, whereas for the larger nanoparticles, their absorption spectra have only been calculated by the FDTD method. Al_n with $n \leq 560$ has two main absorption bands attributed to two kinds of localized surface plasmon resonance modes, vertexes and edges, revealed by the corresponding induced electron densities. RT-TDDFT and FDTD approaches produce obvious spectral differences, and RT-TDDFT predicts more intensive low-energy absorption bands and broader high-energy bands than FDTD. The effect of geometrical relaxation on the absorption spectra of small clusters is visible, which tends to blue shift and broaden the spectra. The smaller the cluster, the larger is the geometrical change.



1. INTRODUCTION

Because of the ability to sustain the localized surface plasmon resonances (LSPR) and the ability to tune LSPR by changing the size, shape, composition, and dielectric environment of nanostructures, noble metal nanoparticles (NPs) have attracted considerable attention and have been applied in many fields such as molecular sensing,[1,2] surface enhanced spectroscopy,[3] dye-sensitized solar cells,[4] and so forth. The noble metal NPs exhibit a strong absorption band in the ultraviolet (UV)−visible light region, which is originated from the collective electron charge oscillations. For example, there is a strong plasmon resonance at ∼520 nm for gold NPs with a diameter D = 22 nm,[5] and the resonance peak is at ∼440 nm for silver spheres with $D \approx 50$ nm.[6] The quasi-static regime holds when the diameter is in the range of approximately 10−50 nm, where the plasmon resonance is nearly independent of the particle size but highly sensitive to the shape of the particle and its embedded media.[7] Unlike most studied silver and gold, the plasmon resonance of aluminum has a wider spectrum region.[8−10] This extended optical response, combined with its natural abundance and low cost, makes aluminum very promising in UV plasmonics applications such as surface-enhanced fluorescence,[11,12] surface-enhanced Raman scattering,[13,14] photovoltaics,[15] and photocatalysis.[16,17]

The modeling and simulation of the optical response of nanostructures provide a detailed, quantitative understanding of these systems, allowing a close interplay of theory and experiment. When the size of the NP is large enough, in which the bulk properties emerge and discrete energy levels turn to form a band structure, the classical electrodynamics methods,[18] such as the discrete dipole approximation,[19] finite difference time domain (FDTD),[20,21] the multiple multipolar method,[22] multiple scattering techniques, transfer matrix approaches,[23] the finiteelement method,[24] and so on can be employed to describe the optical responses. These classical methods have proven to be immensely useful for interpreting a wide range of nanoscience experiments and providing the capability to describe the optical properties of particles up to several hundred nanometers in dimension, with arbitrary particle structures and complex dielectric environments. However, as the size of the metal NPs decreases to the Bohr radius of an exciton, the electronic motion becomes confined and the confinement of the charge carrier discretizes the electronic energy band. As a result, the noble metal NPs with

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the diameter D < 10 nm exhibit molecular-like electronic and optical properties that deviate significantly from their bulk counterparts.[25] Their characteristic plasmon bands are replaced by discrete electronic transitions, which lead to a significant broadening of the plasmon resonance peak. These small size NPs are less well-understood because the above classical methods fail to resolve their responses to the light, but an inevitable quantum mechanical description including full atomistic details is still challenged.

Recently, the linear response time-dependent density functional theory (LR-TDDFT) has been applied to describe the noble metal NPs with a variety of sizes and shapes. For example, Schatz's group used it to investigate Ag_n (n = 10, 20, 35, 56, 84, and 120) tetrahedral NPs;[26] Bae and Aikens[27] and Stener et al.[28] used it to investigate a series of silver and gold octahedra and icosahedra NPs, among which the largest size reaches 309 atoms; Ma et al.[29] combined LR-TDDFT calculation with the exciton model to study the optical properties of coupled Au cluster dimers. However, the LR-TDDFT scheme has its own limits in the practical applications. For instance, it is computationally expensive and even impractical to deal with very complex systems because one has to solve nonordinary eigenvalue equation with a super matrix, which has the dimensions $2N_{occ}N_{vo} \times 2N_{occ}N_{vo}$, where N_{occ} and N_{vo} are the number of occupied and virtual molecular orbitals. An alternative scheme has been developed by numerically solving the time-dependent Kohn–Sham equations directly in time domain,[30−35] here denoted as real-time time-dependent density functional theory (RT-TDDFT). RT-TDDFT scheme shows its advantage on dealing with the excitations of much larger NPs.[36−40] It was shown that the calculations in the time domain can provide us more intuitive information such as the oscillation of electron density than in frequency domain.[41,42] The hybrid schemes,[43−47] which combined classical EM methods and RT-TDDFT, have also been developed to describe the coupling of molecule and plasmon.

Unlike the well-studied Ag and Au NPs, the plasmon excitation of aluminum NPs (ANPs) has not been well-investigated. From a theoretical point of view, Al has only 3 valence electrons, being considered as a good free electron metal. Because of the essential free-electron character of the Al band structure, the interband transitions on the absorption spectrum of Al are less important than in noble metals, and Al can be easily described by the cheap local density approximation with a pseudopotential. However, to our surprise, the first-principle calculations on ANPs are so limited, and the previous studies were focused on very small clusters,[48−50] among which the Al_{13}^- cluster was predicted to be the smallest cluster showing the collective behavior of electronic excitation.[50] In this work, we calculated the absorption spectra of tetrahedral-shaped ANPs with the side lengths ranging from L = 1.41 to 141.4 nm. The size-dependent spectral evolution will thus be demonstrated. For the small clusters with $L \leq$ 3.68 nm (or Al_n with $n \leq$ 560), both TDDFT and FDTD are used to simulate the spectra, which allows us to have a close comparison on the results produced by the quantum and classical simulations. On the one side, we try to calculate the clusters from bottom up using the quantum method; on the other side, we model corresponding NP systems with the classical electrodynamics method to see if the two ends meet. Additionally, we demonstrate how the transition densities and dipoles of corresponding excitations change as the time evolves and assign the electronic excitations in different energy regions to different modes of LSPR.

The paper is organized as follows. Section 2 gives a brief introduction on the theoretical methods and computational details. Section 3 shows the calculated results and analysis. Finally, a concluding remark is given in section 4.

2. MODELING AND COMPUTATIONAL DETAILS

The classical FDTD calculations are performed within the FDTD++ code.[21] A cubic grid with the Yee cell scheme[51] is adopted. The dielectric function of aluminum is described by Drude–Lorentz function with one Drude model term for the conduction band electron motion, adding two Lorentz oscillators for the interband electron transitions[52] as

$$\varepsilon(\omega) = \varepsilon_\infty - \frac{{\omega_D}^2}{\omega(\omega + i\gamma_D)} - \sum_{n=1}^{2} \frac{\Delta\varepsilon_{Ln}{\omega_{Ln}}^2}{\omega(\omega + 2i\delta_{Ln}) - {\omega_{Ln}}^2}.$$

The parameters in the Drude–Lorentz function are obtained by fitting with the experimental permittivity.[53] The fitting parameters and the comparison with experiment permittivity are shown in Table 1 and Figure 1, respectively. The detailed

Table 1. Fitting Parameters

	Drude		Lorentz		
ε_∞	ω_D (eV)	γ_D (eV)	$\Delta\varepsilon_L$	ω_L (eV)	δ_L (eV)
1.00	12.70	0.128	8.86	1.58	0.24
			11.98	2.00	1.18

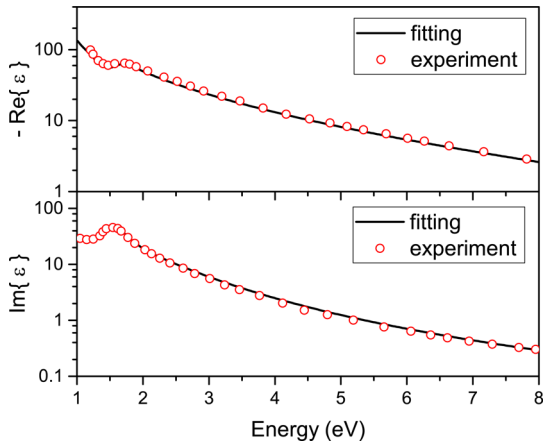


Figure 1. Comparison between the fitting permittivity and the experiment data.

illustration of the FDTD simulation can be found in Figure S1. In all calculations, the light propagation direction is fixed along the +z axis. A cubic simulation box with a side length of 40 nm is chosen for $L \leq$ 3.68 nm NPs, and the grid size is set as 0.16 nm. The larger box and grid size are used with increasing L (see more details in the Supporting Information). The incident light pulse is injected from the plane at z = 10 nm with the functional form, $E(t) = e^{-(t-t_0)^2/\sigma^2} \sin(\omega_0 t)$, where t_0 = 3.0 fs, σ = 0.1 fs, and ω_0 = 3.1 eV. The total simulation time is 60 fs, and the time step size is 2.96×10^{-4} fs.

The tetrahedral clusters are generated by directly cleaving the bulk crystal into tetrahedron shapes. The optimized structures of Al_n with n = 20, 56, 120, 220, 364, and 560 are shown in Figure 2. TDDFT calculations will be performed within the open-source TDDFT code Octopus,[54,55] where the time-dependent Kohn–Sham equation is solved in real space and

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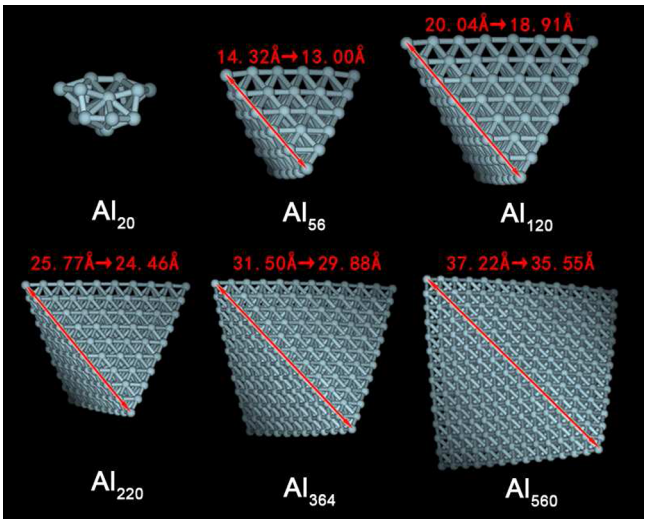


Figure 2. Optimized structures of the aluminum clusters. The numbers inserted shows the corresponding side lengths of clusters before and after the geometric relaxation. The geometrical relaxation makes Al_{20} lose its tetrahedron shape, and the side lengths of all other NPs decrease.

real time. The absorption spectra of both the optimized and unoptimized structures are calculated. The geometrical optimization is finished within Vienna ab initio simulation package.[56,57] Because all NPs are composed of even number of atoms, all electrons are paired, and the spin-restricted DFT method is used. The simulation grid is localized spherically around each atom with a radius of 8 Å, and the spacing between each grid point is 0.3 Å. The Troullier−Martins form pseudopotentials[58] are generated in the general gradient approximation to treat the aluminum atoms and the Perdew−Burke−Ernzerhof functional[57] is used through all calculations. The Fermi-dirac-type electron smearing of 0.02 eV (which has the meaning of thermal temperature) is used when the ground state convergence is hard to reach.

The time propagation is performed using the approximated enforced time-reversal symmetry method,[54] and the fourth order Taylor expansion method is used to approximate the exponential of the Hamiltonian. The time step is set to 1.65×10^{-3} fs ($0.0025\hbar$/eV) and turns out to be very stable for all systems. To get the optical absorption spectrum, a δ-kick is applied to the electron density, and the systems are propagated over a period of 66 fs ($100\hbar$/eV). The absorption cross-section is obtained by performing Fourier transformation of time-dependent induced dipole as

$$S(\omega) = \frac{4\pi\omega}{c\kappa} \mathrm{Im} \int_0^T \{d(t) - d(0)\} e^{i\omega t}\, dt \cdot e^{-\gamma t} \quad (1)$$

where $S(\omega)$ is the absorption cross section, $d(t)$ is the dipole moment of system at time t, κ is the kick strength, and c is the speed of light. Here we use an exponential damping form with γ = 0.128 eV. To be precise, $S(\omega)$ should be given by averaging over all basis directions x, y, and z. However, because all NPs have an approximately tetrahedral symmetry (expect for Al_{20}), if we choose a suitable coordinate system, the difference with respect to the polarization direction will be neglectfully small (see the isotropic tests in the Supporting Information).

To characterize the electronic excitation, we calculate the time-dependent light-induced electron densities of the corresponding electronic excitations and the time-dependent induced dipoles with an applied external laser field, $\vec{E}_{ext}(t) = \vec{E}_{max}\, e^{-(t-t_0)^2/2\tau^2} \cos(\omega_0 t)$. Here $|\vec{E}_{max}|$ is the maximum amplitude of the field and is set to be 0.01 V/Å, which is small enough to keep the response linear. t_0 = 7.26 fs, τ = 1.98 fs, and ω_0 is the corresponding excitation energy of the selected absorption peak. Additionally, the frequency-dependent electron density[38] (transition density[39]) is calculated by the Fourier transform of time-dependent induced density as $\Delta\rho(r,\omega) = \mathrm{Im}\int_0^T \{\rho(r,t) - \rho(r,0)\}\, e^{i\omega t}\, dt\, e^{-\gamma t}$.

3. RESULTS AND DISCUSSION

3.1. Density of States (DOS). The pure tetrahedral Al clusters generated by directly cleaving the bulk crystal may not be stable in realistic experimental environment because of the high chemical activity and bond tension that the outer edge atoms feel. Their geometries are thus optimized by the first-principles method. It is found that the geometry relaxation leads to the structure change more or less. The atoms in the vertexes of the clusters are forced to move toward the core, and in the end, the side lengths of the small clusters shrink, as shown in Figure 2. The larger the cluster, the less is the structure change. The geometry optimization makes the smallest cluster Al_{20} twist and lose its original shape and symmetry. However, all other clusters remain almost the same shape as their unrelaxed structures. It is noted that this geometrical optimization can only find the local energy minimum not the global energy minimum.

On the basis of the optimized structures, we calculated their DOS, as shown in Figure 3. It is noted that the energy levels

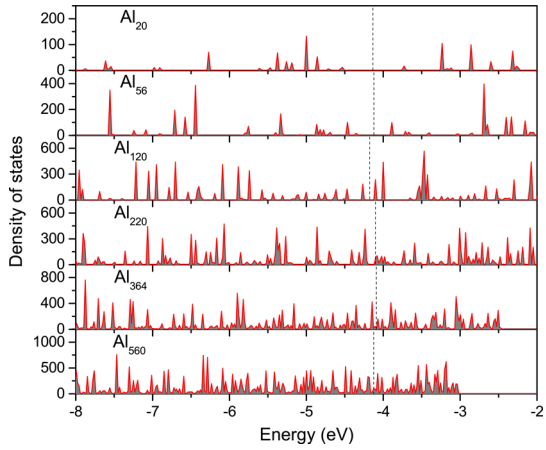


Figure 3. DOS of Al_n with the optimized structures. The Fermi energy levels are marked as the dashed lines.

above −2.5 eV in Al_{364} and −3 eV in Al_{560} regions are missing because we only account for 150 virtual orbitals (VO) in the calculations. Figure 3 shows that the Fermi energy level lying at about −4.2 eV, which is defined at the center between the highest occupied molecular orbitals (HOMOs) and lowest unoccupied molecular orbitals (LUMOs) and is not very sensitive to the cluster size. There is an obvious transition from the molecule-like discrete energy level structure into a bulklike band structure, as the system gets larger and larger. The energy gap between the HOMO and LUMO shrinks rapidly and eventually vanishes, which demonstrates a trend from molecular to metallic transition. Al_{20}, Al_{56}, and Al_{120} are more like molecular systems, whereas Al_{220}, Al_{364}, and Al_{560} tend to be metallic. A similar trend can also be found in previous work,[59]

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where the silver cluster containing 136 atoms still has a band gap in its DOS, whereas the band gap disappears for Ag_{374}.

3.2. Absorption Spectra. To check the accuracy of the RT-TDDFT scheme, at first, we calculate the absorption spectra of Al_{20} by using both RT-TDDFT and LR-TDDFT. The results are shown in the Supporting Information. The two schemes produce nearly the same results in the low-energy region (below 5 eV), but quite different results in the higher energy region, especially when the number of VOs involved in the LR-TDDFT calculation are not enough. We find that the more the VOs are involved, the smaller the deviation. To get the correct result in the high-frequency region, the number of VOs should be 3 times more than the involved occupied orbitals (30 here), which is a challenge for the calculation of large systems. This calculation indicates that the RT-TDDFT method is more computationally efficient for the metal nanoclusters. To characterize the states, we plot the corresponding molecular orbitals, as listed in Figure S4. It can be seen that the electron density is distributed from the bulk to the surface as the energy level goes up, and most of the excitations consist of a number of molecular transitions that are from deep down occupied levels to high above unoccupied levels, which, in a way, can be seen as the future of the surface plasmon.

Figure 4a,b shows the calculated absorption spectra of Al_{56}, Al_{120}, Al_{220}, Al_{364}, and Al_{560} with optimized and unoptimized

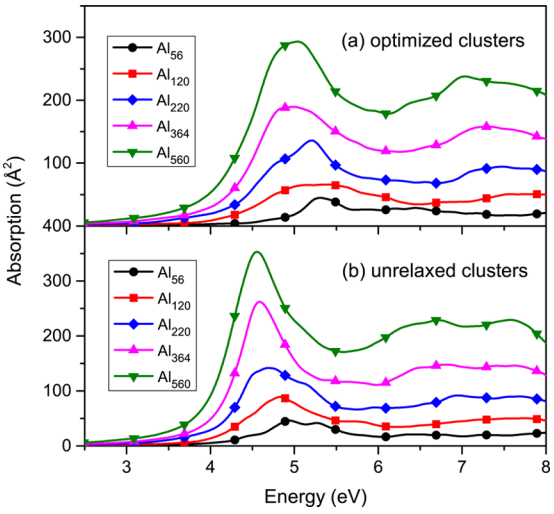


Figure 4. Absorption spectra of Al_n calculated by RT-TDDFT with the (a) relaxed and (b) unrelaxed structures.

structures by the RT-TDDFT scheme. It is clear that their spectra are nearly continuously spanned into a wide-energy region. There are two major absorption bands, an intensive low-energy band lying at 4.0−5.5 eV and a broad high-energy band at 6.0−8.0 eV. As the size increases, the spectrum, especially the first low-energy band, red shifts, and the spectral line becomes smoother and the band width gets narrower. The geometry relaxation tends to broaden the spectra and increase the excitation energies of the first absorption bands because after the geometry relaxation, the cluster structure loses its symmetry and the bond lengths between the neighbor atoms are not uniform, especially those at the surfaces. We thus observed a significant band broadening.

The absorption spectra of equally sized Al clusters have also been calculated by the FDTD scheme, and the corresponding results are shown in Figure 5a. Comparing the spectra of small unrelaxed clusters produced by TDDFT, as shown in Figure 4b,

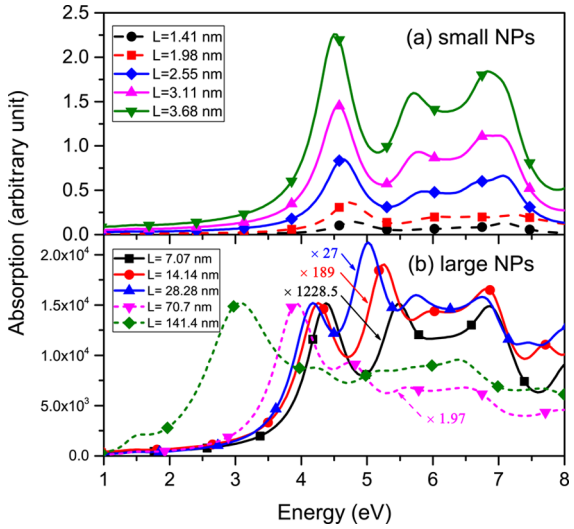


Figure 5. Calculated absorption spectra of NPs by FDTD. (a) Absorption spectra of clusters with side lengths ranging from 1.41 to 3.68 nm. (b) Normalized spectra of larger NPs with side lengths ranging from 7.07 to 141.4 nm.

with those in Figure 5a, we observe that when $L < 2.55$ nm, there is a larger deviation, indicating that the quantum confinement effect is significant in these systems and the classical method cannot correctly predict the result. However, the deviation between the results produced by RT-TDDFT and FDTD reduces as the cluster size increases. Figure 5 shows that as the NP sizes increase, the spectra red shift and the spectral lineshapes of Al NPs change significantly, even for larger NPs. This is different with Ag and Au NPs, in which the plasmon resonance is nearly independent of the particle size when the diameter of the Ag or Au sphere is in the range of approximately 10−50 nm.[7,60]

3.3. Characterizing the Excitations by the Induced Electron Densities and Dipoles. To characterize the absorption peaks, we first show the induced electron density of corresponding excitation in Figure 6. Here, we focus on two of the strongest peaks, one is within the first low-energy absorption band and the other is within the second high-energy band. $\Delta\rho(\omega)$ yields a clear visual insight into the optical response of the LSPR distributed in space. The induced density at the first major peak is mainly localized in the vertexes of the tetrahedron, whereas that in the high-energy band locates on the confronted edges. Both of the bands are generated by LSPR absorption but in two different response modes. These pictures are related to the classical views of plasmon as a near-free electron cloud oscillating between the opposite sides of the NP, which indicates that the system of a few nanometers can sustain the surface plasmon, and its absorption is governed primarily by coherent oscillations of the conduction band electrons. Oppositely, there exists just one collective dipole mode in a spherelike gold cluster.[38,61] In the absence of the d electron screening and the protective ligand effect, here the transition density also emerges from the inside core to the surface of the cluster with increasing size.

Additionally, we characterize the different absorption modes in the time domain. When the oscillating external laser field with a large time duration is applied, the dipole moment

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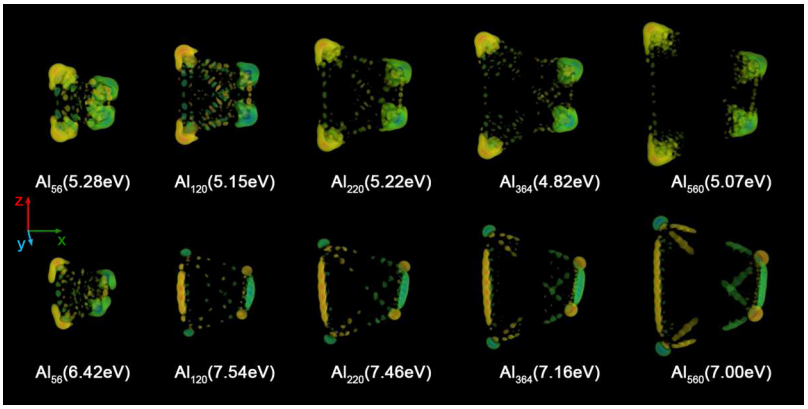


Figure 6. Plots of induced electron densities of the corresponding major peaks. Red and blue colors mean positive and negative values of densities, respectively. The value in the range of −0.01−0.01 is cut off to make the picture clear.

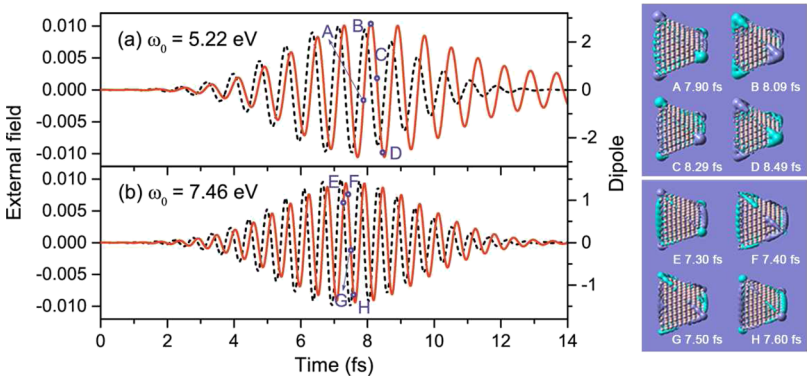


Figure 7. Time evolution of applied external fields (dashed line) and induced dipoles of the system (red line) under the resonance of selected modes of Al_{220} with (a) ω_0 = 5.22 eV and (b) ω_0 = 7.46 eV. The snapshots of time-dependent induced electron densities are shown on the right panels (isovalue = 0.002).

changes with the field as a function of time and shows the response pattern. Here, we take Al_{220} as an example and plot the variation of the external fields and the induced dipoles on the left side of Figure 7. The form of the applied fields here has been introduced in section 2, which is a cosine function in a wide Gaussian package, and the central frequencies correspond to the excitation energies of two major absorption peaks (ω_0 = 5.22 and 7.46 eV). The oscillation of the system dipole keeps in step with the external field and has a slight time lag because of the polarization effect, where the field frequency is not so fast for electron polarization to follow. After the external field fades away, the dipole still keeps oscillating for a long time, but the amplitudes of the two modes are different (see the oscillation in a longer time scale in Figure S5).

Because the strength of the applied field is weak, only the density difference can be recognized, and the snapshots of electron density difference distribution at some selected time points are shown on the right side of Figure 7. The time domain view enable us to see the electron density oscillating around the cluster surface in a more straightforward way, and it shows the same two kinds of excitation modes as from the frequency domain view: the vertex mode in the lower energy band and the edge mode in the higher energy band.

4. CONCLUDING REMARKS

We have performed a detailed theoretical study on the absorption spectra of aluminum NPs with different sizes and demonstrated the size-dependent structures and spectral evolution. It is found that the plasmon resonances of tetrahedral ANPs are highly sensitive to the particle size. As the size of the system increases, the energy level gaps become smaller and discrete molecule-like energy levels start to transform to continuous bulklike bands, and therefore, the low-energy absorption peaks become narrower. Meanwhile, different single-particle excitations merge together, leading to strong collective plasmon excitation peaks. The change of the spectral line shape starts to reduce when the side length of the tetrahedron NP increases to 70 nm. However, we still observe an apparent spectral red shift when L increases from 70 to 141 nm.

The absorption spectra of small aluminum clusters Al_n ($n \leq 560$) show two wide absorption bands at around 4.0−5.5 eV and 6.0−8.0 eV. To characterize the two plasmon bands of small clusters, we plot the corresponding induced electron densities of the two main peaks and find out that the two absorption bands correspond to two kinds of LSPR modes, vertexes and edges, which explain why the relative intensity of the vertex mode becomes weaker and weaker as the cluster size increases, and it eventually disappears when L increases to 70 nm.

For smaller clusters Al_n ($n \leq 560$), FDTD and RT-TDDFT produce quite different spectral line shapes. The spectra calculated by the RT-TDDFT scheme have a broader bandwidth and a relatively intensive low-energy absorption band. This deviation may be attributed to the quantum confinement effect of the small cluster and the inaccurate

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function form of the dielectric constant. In our FDTD simulation, the function of the dielectric constant is obtained by fitting the permittivity of the bulk material, which is definitely unsuitable for small clusters.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.8b02450.

Illustration of the FDTD simulation, isotropic tests on Al_{56}, frequency domain calculation on Al_{20}, and dipole oscillation in a longer time scale (PDF)

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Notes

The authors declare no competing financial interest.

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